

MAT 5403 001

Functional Analysis I

Course Syllabus - Fall 2026



Course Information

Course Description: Prerequisite: MAT 2233, MAT 4273, and MAT 5203, or their equivalents.

Topological vector spaces, inner product spaces, normed spaces, Hilbert spaces and Banach spaces, dual spaces, Hahn-Banach theorem, and bounded linear operators. This course has Differential Tuition. Course Fee: GS01 \$90.

Credit Hours: 3

Course Modality: Traditional in-person

Meeting Times

Duration: 08/19/2026-12/11/2026

Campus: Main Campus

Location: [NPB 1.122](#)

Time(s): MW 6:00 PM - 7:15 PM

Learning Goals

At the end of the course, you will be able to:

- understand the fundamental principles of Functional Analysis and the basic results concerning Operator Theory in Hilbert and Banach spaces.

Communicate with Me

Instructor Name: Gelu Popescu

Department
Mathematics

Office Location
FLN 4.01.28

Student Hours
TR 2:30-3:45 pm (in my office)

Email Address: gelu.popescu@utsa.edu

Preferred Method of Communication

e-mail, office hours

About Me & My Teaching Philosophy

To learn mathematics you need to do mathematics by solving lots of problems. Any new mathematical notion should be supported by examples and counterexamples. All the proofs should be rigorous and the role and impact of the new theorems should be emphasized. Whenever possible, the theory should be anchored to the basic results already known by the students.

Course Materials

Elementary Functional Analysis

Authors: Barbara D. MacCluer

To enhance your learning and adequately prepare for all assignments and assessments, please consider all course materials listed above, along with additional content provided in class, posted to our course site, or included in the Course Schedule, as required reading and study unless indicated as optional.

Additional Course Information

Course Outline:

- **Chapter 1:** Hilbert Space Preliminaries
- **Chapter 2:** Operator Theory Basics
- **Chapter 3:** The Big Three
- **Chapter 4:** Compact Operators
- **Chapter 5:** Banach and C^* -algebras (As time permits).

Assessments and Assignments

Homework assignments will be given in class after each lecture. In order to master the material of the course, it is extremely important that you do your homework. You should make every effort to solve the assigned problems using the concepts learned from the lectures and readings. If you have not practiced the techniques within the homework problems, you will have serious difficulties to work problems on exams. The homework assignments will not be collected and graded. However, any questions related to the homework should be addressed during the office hours.

Activities and Grading

Activity	Quantity	Percentage
Exam 1		30
Total		100%

Activity	Quantity	Percentage
Exam 2		30
Final Exam		40
Total		100%

Distribution of Course Assignments, Their Quantity, and Contribution to Final Grade.

Grade Distribution and Letter Grade

Letter Grade	Grade Range
A+	97.5-100
A	95-97.5
A-	90-95
B+	87.5-90
B	85-87.5
B-	80-85
C+	77.5-80
C	75-77.5
C-	70-75
D+	67.5-70
D	65-67.5
D-	60-65
F	less than 60

Grade Distribution and Letter Grade

Course Guidelines for the Use of Generative Artificial Intelligence

The use of Generative Artificial Intelligence in this course is prohibited.

Course Expectations & Policies

Video and Audio Recording

As the instructor of this course, I may record meetings and lessons. You are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Recordings may not be published, reproduced, or shared with those not in the class. If the instructor or a UTSA office plans any other uses for the recordings, consent of the students identifiable in the recordings is required before such use unless an exception is allowed by law. For more information on your privacy and class recordings, review [Student Privacy \(FERPA\) in Virtual Classrooms and Other Educational Recordings](#) and the [Guide to Secure Video Conferencing Tools](#).

Academic Freedom

Academic freedom¹ is a cornerstone of the University. Academic freedom in its teaching aspect is fundamental for the protection of the rights of the teacher in teaching and of the student to freedom in learning.² Each faculty member is entitled to full freedom in the classroom discussing the subject that the faculty member teaches.³ The University of Texas at San Antonio will not penalize or discipline members of the faculty because of their exercise of academic freedom.

Along with this freedom comes responsibility. It is the responsibility of faculty members to ensure that topics taught are related to the classroom subject. Students should be free to take reasoned exception to the data or views offered in any course of study and to reserve judgment about matters of opinion, but they are responsible for learning the content of any course of study for which they are enrolled.⁴ It is not the proper role of the university or any outside agency to attempt to shield individuals from ideas and opinions they find unwelcome, disagreeable, or even deeply offensive.⁵ Engaging with new ideas and perspectives helps students grow intellectually and is beneficial to the educational process.

1. Statement adapted from Texas A&M University's [Syllabus Statement Regarding Academic Freedom](#)
2. 1940 Statement of Principles on Academic Freedom and Tenure
3. Board of Regents Rule 31004; HOP Policy 4.02
4. American Association of University Professors Joint Statement on Rights and Freedoms of Students
5. The Chicago Statement

Syllabus Changes

The syllabus is subject to change at the instructor's discretion. Any changes/corrections to the course materials, assignment dates, or other updates will be communicated to the students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus.

Course Schedule

For a list of important university-wide dates, review [One Stop's academic calendar](#).

Due Date	Activities/Assignments	Points
9/28/26	Exam 1	100

Due Date	Activities/Assignments	Points
11/9/26	Exam 2	100
12/7/26	Final Exam	100

Department Policies & Resources

Student Advocate

The student advocate is a person who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor; it is highly encouraged that you first reach out to the instructor with any issues to see if they can be resolved.

To reach the department's student advocate, please email mathematics@utsa.edu and include "Student Advocate" in the subject line of your email.

Tutoring Services

UT San Antonio provides many opportunities for student success, including tutoring services for specific math courses. The list below is subject to change based on availability.

MAT 1023, MAT 1043, MAT 1053, and MAT 1073

- We offer peer-to-peer academic support to students enrolled in a variety of courses. Both drop-in tutoring and one-on-one appointments are available.
- **Location:** [Tomás Rivera Center for Academic Excellence](#) MS 2.02.18 (behind the Rowdy statue)

MAT 1053 and MAT 1073

- Synergistic Targeted Academic Review Sessions (STARS), peer-led, interactive, and group focused sessions which cover traditionally difficult topics. The schedule will be provided by instructors through Canvas.
- **Location:** Determined by STARS tutors

MAT 1093, MAT 1213, MAT 1223, MAT 2213, MAT 2253, MAT 1313

- The Math Gym is unique in that it's a WeBWork lab designed to offer our STEM core students support from math majors and minors through targeted SI sessions. The schedule for drop-in hours and course-specific targeted sessions will be provided by instructors through Canvas.
- **Location:** [Math Gym, Main Building 0.414](#)

Note: Course-specific support for upper-division courses may be available for a particular semester. Please request this information from your instructor.

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) to find important and valuable resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, accommodations for pregnant students, campus safety and emergency preparedness, and the Roadrunner Creed.
- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support](#) page.
- UT San Antonio provides reasonable accommodations to students via [Student Disability Services](#). For more details on eligibility, policies, and requirements, please visit [Applying for Accommodations](#) or call (210) 458-4157.
- Your well-being is important. In addition to academic growth, life as a student can include many emotional experiences and opportunities. Being mindful of your own needs and available support can provide you with the tools to successfully navigate these experiences. Student support for well-being, including [24/7 mental health services](#), can be accessed from the [Well-being at UT San Antonio](#) site for students. If needed, we can schedule a meeting, and I can connect you to resources. You are not alone; I am available to help you find resources that might best meet your needs.
- [Student Assistance Services \(SAS\)](#) is available to help you navigate academic and personal challenges. Campus Advocates in SAS are a central point of contact for students seeking clarity on university processes and policies. They also provide a direct connection to helpful resources and advocacy rooted in care. Campus Advocates offer confidential support and personalized guidance to help students succeed.
- The [Roadrunner Pantry](#) is here to help UT San Antonio students stay healthy and focused by providing free access to groceries, toiletries, and other essentials. With locations at the Main and Downtown campuses, the pantry is organized like a small store and offers items such as canned goods, snacks, and hygiene products. Students with a valid UTSA ID can visit the pantry to pick up what they need. Our goal is to ensure every Roadrunner feels supported and has the resources to succeed. If you're ever short on food or supplies, the Roadrunner Pantry is a welcoming space you can count on.
- Students at UT San Antonio are responsible for ensuring their work is consistent with UT San Antonio's standards for academic integrity. Students should review [Section 203 of the UT San Antonio Student Code of Conduct](#) for appropriate standards of academic integrity.
- UT San Antonio provides numerous services for students from counseling to tutoring to a food pantry. Visit [Student Affairs Programs and Services](#) and [Student Success](#) for more information.
- Visit the [UT San Antonio Libraries and Museums](#) site for access to journals, research tutorials, and tech gear you can borrow and to find your department's librarian.
- Enroll in the [Roadrunner Success Playbook](#), an open-enrollment, self-paced, online hub in Canvas tailored to ensure you have the resources you need to excel at UT San Antonio.
- Follow [Digital Learning Netiquette](#) standards for your online communication activities.